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Electrical lead-through and electronics housing

Abstract

An electrical lead-through for an electronics housing, comprises a first stop element with at least one first through-hole, a second stop element with at least one second through-hole, an elastic element which extends between the first stop element and the second stop element along a first axis, and at least one electrical contact element with a first end and a second end lying opposite the first end. The first end of the contact element is movably arranged in the first through-hole and the second end is movably arranged in the second through-hole. The elastic element is suitable for deforming when the first stop element and the second stop element are moved relative to each other, and the first stop element and the second stop element are each suitable for being fixed to the electronics housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is related to and claims the priority benefit of DPMA Patent Application No. 10 2021 122 931.6, filed on Sep. 6, 2021, and International Patent Application No. PCT/EP2022/072052, filed Aug. 5, 2022, the entire contents of which are incorporated herein by

reference.

TECHNICAL FIELD

(2) The invention relates to an electrical lead-through and to an electronics housing.

BACKGROUND

(3) Electronics housings are used in various environments in the industry. If an electronics housing is to be used in an environment that is exposed to explosion hazards, the electronics housing must have a pressure-resistant casing in order to prevent a flame or a spark from escaping from the electronics housing.

(4) Pressure-resistant casings are particularly difficult to realize in the region of the electrical lead-through on the housing, i.e., at the electrical interface with the electronics arranged in the housing. So far, electrical lead-throughs have been cast in a complicated manner, which is associated with expensive materials and a long drying time. The production costs and the production time of an electronics housing are thus increased.

SUMMARY

(5) It is therefore an object of the invention to provide an electrical lead-through which optimizes the manufacturing costs and the manufacturing time of an electronics housing.

(6) This object is achieved by the electrical lead-through for an electronics housing according to the present disclosure.

(7) The electrical lead-through according to the invention comprises: a first stop element with at least one first through-hole, a second stop element with at least one second through-hole, an elastic element which extends between the first stop element and the second stop-element along a first axis, at least one electrical contact element with a first end and a second end lying opposite the first end. The first end is movably arranged in the first through-hole and/or the second end is movably arranged in the second through-hole. The elastic element is suitable for deforming when the first stop element and the second stop element are moved relative to each other. The first stop element and the second stop element are each suitable for being fixed to the electronics housing.

(8) The electrical lead-through according to the invention allows for providing an electrical lead-through for an electronics housing which, even in the event of high temperature fluctuations causing a contraction of the material, reliably seals the electronics housing towards the outside at its electrical interface. Moreover, the electrical lead-through according to the invention allows for a simple and convenient installation in an electronics housing.

(9) According to one embodiment of the invention, the first through-hole of the first stop element has a taper with a first hole diameter, and the first end of the electrical contact element has a taper with a first diameter. The first hole diameter and the first diameter are the same, and/or the second through-hole of the second stop element has a taper with a second hole diameter, and the second end of the electrical contact element has a taper with a second diameter. The second hole diameter and the second diameter are the same.

(10) According to one embodiment of the invention, the first stop element extends transversely to the first axis and the second stop element extends transversely to the first axis.

(11) According to one embodiment of the invention, the second stop element comprises a first fastening unit with an adhesive surface, a welding surface, a thread and screw, a groove and a snap ring, or a hole and a rivet.

(12) According to one embodiment of the invention, the elastic element extends longitudinally to the first axis and has a cross-sectional shape that is at least partially conical.

(13) According to one embodiment of the invention, the elastic element is made from elastomer.

(14) The above object is further achieved by an electronics housing for enclosing an electronic unit.

(15) The electronics housing according to the invention comprises: an electrical lead-through according to the invention, a housing body.

(16) The housing body has a connection channel for receiving the electrical lead-through. The connection channel has an inner wall which is partially complementary to an outer wall of the

elastic element, so that the connection channel is sealed by the electrical lead-through when the electrical lead-through is arranged in the connection channel. The connection channel has a cavity which is arranged such that, when the first stop element and the second stop element are moved relative to each other, the elastic element can be deformed into the cavity. The connection channel has a first fastening region which is suitable for being fixed to the first stop element, and the housing body has a second fastening region which is suitable for being fixed to the second stop element, so that, when the first fastening region is fixed to the first stop element and the second fastening region is fixed to the second stop element, the elastic element is deformed into the cavity.
(17) According to one embodiment of the invention, the second fastening region comprises an adhesive surface, a welding surface, a thread, a groove for a snap ring, or a hole for a rivet, and the second stop element comprises a first fastening unit with an adhesive surface, a welding surface, a thread and screw, a groove and a snap ring, or a hole and a rivet.
(18) According to one embodiment of the invention, the connection channel extends along a second axis, and the first fastening region and the second fastening region are spaced apart along the second axis by a first distance. The first stop element and the second stop element of the electrical lead-through are spaced apart along the first axis by a second distance. The first distance is smaller than the second distance.
(19) According to one embodiment of the invention, the electrical contact element has a first stop at the taper at its first end, and the first stop element has a second stop at the first through-hole, and/or the electrical contact element has a third stop at the taper at its second end, and the second stop element has a fourth stop at the second through-hole, so that, when the first fastening region is fixed to the first stop element and when the second fastening region is fixed to the second stop element, the first stop is in contact with the second stop and/or the third stop is in contact with the fourth stop.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present disclosure is explained in more detail on the basis of the following description of the figures. In the figures:
(2) FIG. **1** shows a schematic sectional drawing of an electrical lead-through according to the present disclosure which is installed in an electronics housing,
(3) FIG. **2** shows the electrical lead-through of FIG. **1**, which is fastened in the electronics housing,
(4) FIG. **3** shows a sectional drawing of the electrical lead-through shown in FIG. **1** with an electronics housing,
(5) FIG. **4** shows an enlarged representation of two details of FIG. **3**.
DETAILED DESCRIPTION
(6) FIG. **1** shows an electronics housing **10** according to the invention with a housing body **11** and an electrical lead-through **20**.
(7) The electrical lead-through **20** comprises a first stop element **21**, a second stop element **23**, an elastic element **25** and an electrical contact element **26**.
(8) The first stop element **21** has at least one first through-hole **22**. The first through-hole **22** is suitable for receiving an electrical contact element **26**. In the embodiments shown in FIGS. **1** to **4**, the first stop element **21** has three through-holes in order to receive an electrical contact element in each through-hole. The first stop element **21** is suitable for being fixed to the electronics housing **10**.
(9) The second stop element **23** has at least one second through-hole **24**. The second through-hole **24** is suitable for receiving an electrical contact element **26**. In the embodiments shown in FIGS. **1** to **4**, the second stop element **23** has three through-holes in order to receive an electrical contact

element in each through-hole. The second stop element **23** is suitable for being fixed to the electronics housing **10**. The second stop element **23** has a first fastening unit **29**. The first fastening unit **29** is, for example, an adhesive surface, a welding surface, a thread with a screw, a groove with a snap ring or a hole with a rivet.
(10) The first stop element **21** and the second stop element **23** are made of an electrically insulating material, for example plastic. The first stop element **21** preferably extends transversely to the first axis A**1** and the second stop element **23** preferably extends transversely to the first axis A**1**.
(11) The elastic element **25** is arranged such that it extends between the first stop element **21** and the second stop element **23** along a first axis A**1**. The first stop element **21** and the second stop element **23** thus delimit the elastic element **25**. The elastic element **25** has a first height H**1** along the first axis A**1**. The elastic element **25** has an outer wall **30**. The elastic element **25** is suitable for deforming when the first stop element **21** and the second stop element **23** are moved relative to each other. The elastic element **25** has at least one through-opening in which the at least one electrical contact element **26** is located. The elastic element **25** is arranged such as to be movable relative to the electrical contact element **26**.
(12) The elastic element **25** preferably has an at least partially conical cross-sectional shape longitudinally to the first axis A**1**. The conical cross-sectional shape allows for safe and comfortable installation of the electrical lead-through **20** in the electronics housing **10**. Moreover, if the tip of the cone is directed towards the outside, a conical cross-sectional shape makes it possible for the cone to become wedged in the connection channel **12**, i.e., to seal the connection channel **12**, if there is excess pressure in the electronics housing **10**. Of course, the elastic element **25** can also have other shapes, such as a cylindrical shape, for example.
(13) For example, the elastic element **25** is made of elastomer or comprises an elastomer. Of course, the elastic element **25** can also be made of other elastic materials, provided that the elasticity of the elastic element **25** enables a deformation of the elastic element **25** such that a bead can be formed when the first stop element **21** and the second stop element **23** are pressed together (see FIG. **2**).
(14) The electrical contact element **26** has a first end **27** and a second end **28** opposite the first end **27**. The first end **27** is arranged in the first through-hole **22** and the second end **28** is arranged in the second through-hole **24**. The electrical contact element **26** is made of an electrically conductive material. For example, the electrical contact element **26** is made of copper, silver or gold or has a coating of silver, gold or platinum.
(15) FIG. **4** shows two marked details from FIG. **3** in an enlarged representation. As shown in FIG. **4**, the first through-hole **22** of the first stop element **21** has a taper with a first hole diameter L**1**, and the first end **27** of the electrical contact element **26** has a taper with a first diameter D**1**. The first hole diameter L**1** and the first diameter D**1** are substantially the same. Here, “the same” shall be understood to mean that the electrical contact element can be inserted just barely into the stop element. Alternatively or complementarily, the second through-hole **24** of the second stop element **23** has a taper with a second hole diameter L**2**, and the second end **28** of the electrical contact element **26** has a taper with a second diameter D**2**. The second hole diameter L**2** and the second diameter D**2** are the same.
(16) The first end **27** of the electrical contact element **26** is movably arranged in the first through-hole **22** along the first axis A**1** and/or the second end **28** is movably arranged in the second through-hole **24** along the first axis A**1**. As a result, the electrical contact element **26** facilitates a compression of the first stop element **21** and the second stop element **23**.
(17) FIG. **1** to FIG. **3** show the housing body **11**, which has a connection channel **12** for receiving the electrical lead-through **20**. The connection channel **12** has an inner wall **13** which is partially complementary to the outer wall **30** of the elastic element **25**.
(18) The connection channel **12** has a cavity **14** which is arranged such that, when the first stop element **21** and the second stop element **23** are moved relative to each other, the elastic element **25**

can be deformed into the cavity **14**. The cavity **14** preferably extends around the first axis A**1**, so that the cavity **14** extends concentrically around the elastic element **25** when the latter is arranged in the connection channel **12**.
(19) The connection channel **12** has a first fastening region **15** which is suitable for being fixed to the first stop element **21**. The first fastening region **15** is preferably a shoulder which fixes the first stop element **21**, so that the first stop element **21** is fixed in a direction along the first axis A**1**. In other words, the first stop element **21** is thus held in the connection channel **12**. An exchange of the electrical lead-through **20** is thus easily possible. As an alternative to a shoulder extending substantially orthogonally to the longitudinal axis A**1**, i.e., forming a taper of the connection channel **12** (see FIGS. **1-4**), the shoulder can also extend obliquely to the longitudinal axis A**1**, i.e., can have a conical shape. A conical shape of the shoulder allows the first stop element **21** and the shoulder to be wedged.
(20) According to an alternative embodiment, the first fastening region **15** is glued, welded, for example ultrasonically welded, or connected to the connection channel **12** by means of a mechanical unit, for example a clamping ring.
(21) The housing body **11** has a second fastening region **16** which is suitable for being connected to the second stop element **23**, so that, when the first fastening region **15** is connected to the first stop element **21** and the second fastening region **16** is connected to the second stop element **23**, the elastic element **25** is deformed into the cavity **14**. The second fastening region **16** comprises an adhesive surface, a welding surface, a thread, a groove for a snap ring or a hole for a rivet.
(22) The connection channel **12** extends along a second axis A**2**. Both the first fastening region **15** and the second fastening region **16** are spaced apart along the second axis A**2** by a first distance B**1**. The first stop element **21** and the second stop element **23** of the electrical lead-through **20** are spaced apart along the first axis A**1** by a second distance B**2**. The first distance B**1** is smaller than the second distance B**2**.
(23) As can be seen in FIG. **4**, the electrical contact element **26** has a first stop **31** at the taper at its first end **27**, and the first stop element **21** has a second stop **32** at the first through-hole **22**. Alternatively or complementary, the electrical contact element **26** has a third stop **33** at the taper at its second end **28**, and the second stop element **23** has a fourth stop **34** at the second through-hole **24**. As a result, when the first fastening region **15** is fixed to the first stop element **21** and when the second fastening region **16** is fixed to the second stop element **23**, the first stop **31** is in contact with the second stop **32** and/or the third stop **33** is in contact with the fourth stop **34**.
(24) Hereinafter, the installation of the electronics housing **10**, i.e., installation of the electrical lead-through **20** in the housing body **11**, is described.
(25) First, the electrical lead-through **20** is installed in the connection channel **12** of the housing body **11** and the first stop element **21** is fixed to the first fastening region **15**. Fixing means, for example, pressing, gluing or welding the first stop element **21** to the first fastening region **15**, for example a shoulder. Of course, the first stop element **21** can also be fixed to the first fastening region **15** in some other way, for example by means of a clamping ring.
(26) The second stop element **23** is then pressed along the first axis A**1** in the direction of the first stop element **21**, so that the elastic element **25** is deformed and forms a bead which extends into the cavity **14** of the connection channel **12**. In this position, the elastic element **25** has a second height H**2**, which is lower than the first height H**1** (see FIG. **2**).
(27) Next, the second stop element **23** is fixed to the second fastening region **16** of the housing body **11**. Fixing means, for example, gluing, screwing, riveting or welding the second stop element **23** to the second fastening region **16**. Of course, the second stop element **23** can also be fixed to the second fastening region **16** in some other way, for example by means of a clamping ring.
(28) Thanks to the bead formation on the elastic element **25**, the electrical lead-through **20** allows optimal sealing of the connection channel **12** even in the event of temperature fluctuations. If, for example, the elastic element **25** contracts at low temperatures, this means that the bead is somewhat

reduced but a first gap zone Z**1** and a second gap zone Z**2** is still kept minimal or avoided (see FIG. **2**).

Claims

1. An electrical lead-through for an electronics housing, comprising: a first stop element with a first through-hole; a second stop element with a second through-hole; an elastic element which extends between the first stop element and the second stop element along a first axis; and at least one electrical contact element with a first end and a second end lying opposite the first end, wherein the first end of the electrical contact element is movably arranged in the first through-hole and/or the second end of the electrical contact element is movably arranged in the second through-hole, wherein the elastic element is suitable for deforming when the first stop element and the second stop element are moved relative to each other, wherein the first stop element and the second stop element are each suitable for being fixed to the electronics housing, and wherein the first through-hole of the first stop element has a taper with a first hole diameter, the first end of the electrical contact element has a taper with a first diameter, and the first hole diameter and the first diameter are the same, and/or wherein the second through-hole of the second stop element has a taper with a second hole diameter, the second end of the electrical contact element has a taper with a second diameter, and the second hole diameter and the second diameter are the same.
2. The electrical lead-through according to claim 1, wherein the first stop element extends transversely to the first axis and the second stop element extends transversely to the first axis.
3. The electrical lead-through according to claim 1, wherein the second stop element includes a first fastening unit with an adhesive surface, a welding surface, a thread and screw, a groove and a snap ring, or a hole and a rivet.
4. The electrical lead-through according to claim 1, wherein the elastic element extends longitudinally to the first axis and has a cross-sectional shape that is at least partially conical.
5. The electrical lead-through according to claim 1, wherein the elastic element is made of elastomer.
6. An electronics housing for enclosing an electronic unit, comprising: an electrical lead-through, including: a first stop element with a first through-hole; a second stop element with a second through-hole; an elastic element which extends between the first stop element and the second stop element along a first axis; and at least one electrical contact element with a first end and a second end lying opposite the first end, wherein the first end of the electrical contact element is movably arranged in the first through-hole and/or the second end of the electrical contact element is movably arranged in the second through-hole, wherein the elastic element is suitable for deforming when the first stop element and the second stop element are moved relative to each other, and wherein the first stop element and the second stop element are each suitable for being fixed to the electronics housing; and a housing body having a connection channel for receiving the electrical lead-through, wherein the connection channel has an inner wall which is partially complementary to an outer wall of the elastic element so that the connection channel is sealed by the electrical lead-through when the electrical lead-through is arranged in the connection channel, wherein the connection channel has a cavity which is arranged such that, when the first stop element and the second stop element are moved relative to each other, the elastic element can be deformed into the cavity, and wherein the connection channel has a first fastening region which is suitable for being fixed to the first stop element, and the housing body has a second fastening region which is suitable for being fixed to the second stop element so that, when the first fastening region is fixed to the first stop element and the second fastening region is fixed to the second stop element, the elastic element is deformed into the cavity.
7. The electronics housing according to claim 6, wherein the second fastening region includes an adhesive surface, a welding surface, a thread, a groove for a snap ring, or a hole for a rivet, and the

second stop element comprises a first fastening unit with an adhesive surface, a welding surface, a thread and screw, a groove and a snap ring, or a hole and a rivet.

8. The electronics housing according to claim 7, wherein the connection channel extends along a second axis, and the first fastening region and the second fastening region are spaced apart along the second axis by a first distance, and wherein the first stop element and the second stop element of the electrical lead-through are spaced apart along the first axis by a second distance that is larger than the first distance is smaller than the second distance.

9. The electronics housing according to any one of claim 8, wherein the electrical contact element has a first stop at the taper at its first end, and the first stop element has a second stop at the first through-hole, and/or the electrical contact element has a third stop at the taper at its second end, and the second stop element has a fourth stop at the second through-hole so that, when the first fastening region is fixed to the first stop element and when the second fastening region is fixed to the second stop element, the first stop is in contact with the second stop and/or the third stop is in contact with the fourth stop.